

INFOSYS SPRINGBOARD
SMART URBAN MOBILITY
& TRAFFIC INTELLIGENCE DASHBOARD
TEAM PROJECT DOCUMENTATION

Microsoft Power BI Desktop

TEAM D

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Smart Urban Mobility & Traffic Intelligence | Consolidated Documentation

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1. Introduction and Project Objective

The Smart Urban Mobility and Traffic Intelligence Dashboard is a Power BI-based analytics project developed to convert urban transportation records into an interactive decision-support dashboard. The project combines data preparation, dimensional modeling, DAX calculations, geospatial analysis, route and transit performance, demand intelligence, mobility-mode analysis, environmental sensitivity and time-based demand analysis.

The work was developed progressively across three milestones. Milestone 1 established the analytical foundation through data preparation, validation, a reusable star schema, DAX measures, slicers and a baseline dashboard. Milestone 2 extended the same model into demand, geospatial, route and transit-performance intelligence. Milestone 3 moved further into demand intelligence, pricing proxies, mobility-mode analysis, regional availability and forecasting exploration.

1.1 Project Objective

- Prepare and validate the mobility dataset for reliable analysis.
- Build a reusable star-schema semantic model in Power BI.
- Create dynamic DAX measures that respond to filter context.
- Analyze traffic, congestion, route performance and geographic demand.
- Extend the project into demand intelligence, mobility modes, environmental sensitivity and time-based analysis.
- Present the results through a professional, interactive dashboard.

1.2 Overall Analytical Flow

1. Prepare and validate mobility data.
2. Build FactMobilityTable and supporting dimensions.
3. Create and validate DAX measures.
4. Progress from basic traffic monitoring to geospatial and transit intelligence.
5. Extend the model into demand intelligence and forecasting exploration.

2. Tools and Technologies Used

Tool / Technology	Use in the Project
Microsoft Power BI Desktop	Data model, DAX, dashboards, maps, charts, slicers and forecasting exploration.
Power Query	Data cleaning, data-type validation and transformation.
DAX	Dynamic KPIs, reliability, disruption logic, demand measures and derived indicators.
Microsoft Excel	Source dataset preparation, validation and structured mobility records.
Data Modeling	Fact-dimension relationships and filter propagation using a star schema.
Data Visualization	KPI cards, line charts, column/bar charts, donut charts, maps, tables and scatter plots.

3. Milestone 1 - Data Integration, Data Modeling and Dashboard Foundation

Milestone 1 established the core analytical foundation of the Smart Urban Mobility and Traffic Intelligence project. The main focus was to prepare the 300-record mobility dataset, validate identifiers and data types, create the fact and dimension structure, configure relationships, create reusable DAX measures and build an interactive baseline dashboard.

3.1 Dataset Overview

The project uses a 300-record mobility dataset. The central mobility table contains record-level transportation observations and connects to date, location, route, vehicle and weather dimensions through business keys. The model was designed to support both high-level KPIs and detailed route analysis.

Key Field	Role / Description
Mobility_ID	Unique identifier for each mobility record.
DateKey	Numeric key used to relate mobility records to the date dimension.
Timestamp	Date and time of the recorded observation.
LocationKey	Key linking the record to location attributes.
Route_ID	Route identifier for route-level analysis.
WeatherKey	Key linking the record to weather conditions.
VehicleKey	Key linking the record to vehicle attributes.
Hour / Peak_Period	Used for hourly and peak-period analysis.
Vehicle_Count / Passenger_Count	Observed traffic and passenger quantities.
Avg_Speed_kmph / Delay_Minutes	Core performance measurements.
Congestion_Index / Utilization_Rate	Operational and resource indicators.

3.2 Data Cleaning and Validation

- Identifier fields were retained as text where appropriate; DateKey was treated as a whole-number key and Timestamp as Date/Time.
- Counts and measurements were assigned numeric data types.
- Mobility_ID was treated as the unique record identifier.
- DateKey, LocationKey, Route_ID, WeatherKey and VehicleKey were validated as relationship keys.
- Categorical values were standardized sufficiently for slicers and grouped visuals.
- Dimension keys were checked so that the dimensions could sit on the one-side of one-to-many relationships.

3.3 Data Modeling - Star Schema

The Power BI model follows a star-schema structure. FactMobilityTable stores the measurable mobility events, while the dimension tables provide reusable descriptive context. Single-direction one-to-many relationships were used to keep filter propagation predictable and reduce ambiguity.

From	To	Key	Cardinality	Filter Direction
Dim_Date	FactMobilityTable	DateKey	1 : *	Single
Dim_Location	FactMobilityTable	LocationKey	1 : *	Single
Dim_Route	FactMobilityTable	Route_ID	1 : *	Single
Dim_Vehicle	FactMobilityTable	VehicleKey	1 : *	Single
Dim_Weather	FactMobilityTable	WeatherKey	1 : *	Single

Suggested caption: Figure 1. Power BI Model View showing FactMobilityTable and the five dimension tables.

3.4 Dimension Tables

Dimension	Important Fields	Purpose
Dim_Date	Date, DateKey, Day_Name, Month, Quarter, Year, Is_Holiday, Is_Weekend, Is_Working_Day	Time-based analysis and date filtering.
Dim_Location	City, State, Zone, Location, Latitude, Longitude, Population, Population_Density_per_km2, Road_Class and nearby-facility fields	Geographic and location-level analysis.
Dim_Route	Route_ID, Route_Type, Distance_km, Lane_Count, Speed_Limit_kmph, Capacity_Vehicles_per_Hour, Origin_LocationKey, Destination_LocationKey, Public_Transport_Corridor, Toll_Route	Route characteristics and performance analysis.
Dim_Vehicle	VehicleKey, Vehicle_Type, Typical_Capacity, Ownership_Type, Fuel_Consumption_L_per_100km, Emission_Factor	Vehicle mix, capacity and environmental context.
Dim_Weather	WeatherKey, Weather, Temperature_C, Humidity_pct, Rainfall_mm, Wind_Speed_kmph, Traffic_Impact_Level	Weather context and weather-to-congestion analysis.

3.5 Key DAX Measures

Measure	DAX / Logic	Purpose
Total Vehicles	SUM(FactMobilityTable[Vehicle_Count])	Total traffic volume.
Total Passengers	SUM(FactMobilityTable[Passenger_Count])	Total passengers carried.

Average Speed	AVERAGE(FactMobilityTable[Avg_Speed_kmph])	Average observed speed.
Average Delay	AVERAGE(FactMobilityTable[Delay_Minutes])	Average delay.
Average Congestion	AVERAGE(FactMobilityTable[Congestion_Index])	Average congestion level.
Average Travel Time	AVERAGE(FactMobilityTable[Travel_Time_Minutes])	Average travel duration.
Average Utilization	AVERAGE(FactMobilityTable[Utilization_Rate])	Average utilization.
Average Wait Time	AVERAGE(FactMobilityTable[Avg_Wait_Minutes])	Average waiting time.
High Congestion %	COUNTROWS(FILTER(FactMobilityTable,[Congestion_Index] >= 60)) / COUNTROWS(FactMobilityTable)	Share of high-congestion observations.
Peak Hour Vehicles	MAXX(VALUES(FactMobilityTable[Hour]), [Total Vehicles])	Highest hourly vehicle volume.
Passengers per Vehicle	DIVIDE([Total Passengers],[Total Vehicles],0)	Passenger-to-vehicle ratio.
Distinct Routes	DISTINCTCOUNT(FactMobilityTable[Route_ID])	Number of unique routes represented.
Route Efficiency	DIVIDE([Average Speed],[Average Travel Time],0)	Derived route efficiency indicator.

3.6 Baseline Dashboard and Visuals

The baseline dashboard used four KPI cards, interactive slicers, analytical charts and a route-level detail table. The main baseline values documented for the full selected range were Total Vehicles = 226K, Total Passengers = approximately 1M, Average Speed = 29.59 km/h and Average Delay = 24.09 minutes.

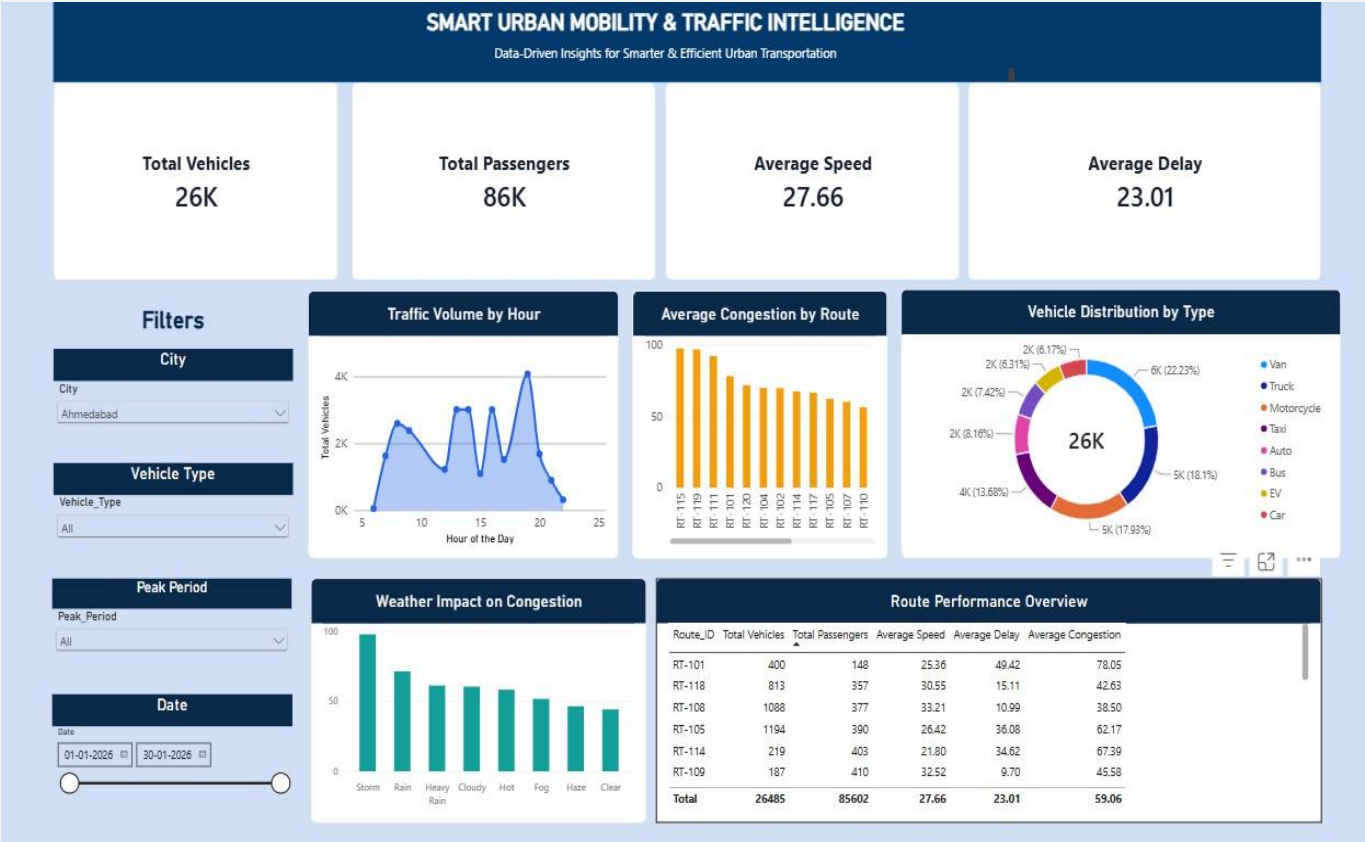
Visual	Purpose
Total Vehicles / Total Passengers / Average Speed / Average Delay KPI cards	Immediate executive summary.
Traffic Volume by Hour - Line Chart	Identify hourly peaks, dips and fluctuations.
Average Congestion by Route - Column Chart	Compare congestion across route IDs.
Vehicle Distribution by Type - Donut	Show composition of vehicle activity.
Weather Impact on Congestion - Column Chart	Compare average congestion across weather categories.
Route Performance Overview - Table	Provide exact route-level vehicles, passengers, speed, delay and congestion.

Interactive slicers included City, Vehicle Type, Peak Period and Date. Slicer selections propagate through the model so the DAX measures recalculate in the active filter context.

3.7 Milestone 1 Key Insights

- Traffic volume changes substantially by hour, allowing high- and low-activity periods to be identified.
- Congestion is not uniform across routes, so route-level comparison can reveal corridors requiring closer attention.
- Vehicle activity is distributed across multiple vehicle types, providing a clear view of traffic composition.
- Average congestion varies across weather categories, supporting further investigation of external-condition effects on traffic.
- The route table complements visual summaries by exposing exact route-level values for vehicles, passengers, speed, delay and congestion.

SCREENSHOT - MILESTONE 1 FINAL DASHBOARD



4. Milestone 2 - Demand, Geospatial and Transit Intelligence

Milestone 2 extended the Milestone 1 dashboard from basic monitoring into deeper demand, geospatial, route and transit-performance analysis. The same core dataset, semantic model and star-schema relationships were retained, and a second report page was built rather than creating a separate project.

4.1 Objectives

- Identify high- and low-demand locations using passenger and trip volume.
- Compare route performance using speed and supporting operational metrics.
- Observe transit performance over time to identify variation and peaks.
- Enable geographic drill-down from City to Zone to Location.
- Provide KPI cards for trips, passengers, reliability and disruptions.
- Preserve the Milestone 1 model while adding purpose-built Milestone 2 calculations.

4.2 Milestone 2 Analytical Workflow

Step	Process	Implementation
1	Model validation	Confirmed Milestone 1 fact/dimension structure and relationships.
2	DAX measures	Created Total Trips, Transit Reliability, Disruptions and a derived Passenger Satisfaction index; reused Total Passengers.
3	Slicers	Added Date, Time, Route, Location/City and Weather controls.
4	Demand analysis	Built a passenger-demand map using latitude, longitude and Total Passengers as bubble size.
5	Route performance	Built route-wise bar chart using Route_ID and Average Speed with supporting tooltips.
6	Trend analysis	Built line chart of Average Speed by Date.
7	Geographic drill-down	Built City -> Zone -> Location hierarchy.
8	Operational detail	Added route performance details table.
9	Flow-map preparation	Prepared MobilityFlow support table with ordered coordinates for origin-destination analysis.

4.3 DAX Measures and Definitions

Measure	Logic / Definition	Purpose
Total Trips	COUNTROWS(FactMobilityTable)	Counts mobility records as trip-volume proxy.
Total Passengers	SUM(FactMobilityTable[Passenger_Count])	Aggregate passenger demand.
Transit Reliability %	Trips with Delay_Minutes <= 5 / Total trips	Share of trips meeting the defined reliability threshold.
Disruptions	Count of records with Delay_Minutes > 10	Operational stress indicator.
Passenger Satisfaction %	Derived from average delay using a bounded analytical index	Service-quality proxy; not survey-measured satisfaction.

4.4 Dashboard Visuals

Visual	Configuration	Purpose
Passenger Demand by Location	Latitude + Longitude + Total Passengers as bubble size	Show where passenger demand is geographically concentrated.

Route-wise Performance	Route_ID + Average Speed; Trips, Passengers, Delay and Congestion in tooltips	Compare route performance while retaining context.
Transit Performance Over Time	Date + Average Speed	Show performance variation over the reporting period.
Geographical Drill-Down	City -> Zone -> Location; Total Trips as bubble size	Navigate from broad geography to detailed locations.
Route Performance Details	Route_ID, Total Trips, Total Passengers, Average Speed, Average Delay, Average Congestion	Provide exact route-level values behind the charts.

The page used Date, Time, Route, Location/City and Weather slicers. The dashboard was validated so that KPI cards, charts, maps and the detail table responded to filter selections.

SCREENSHOT - MILESTONE 2 FINAL DASHBOARD

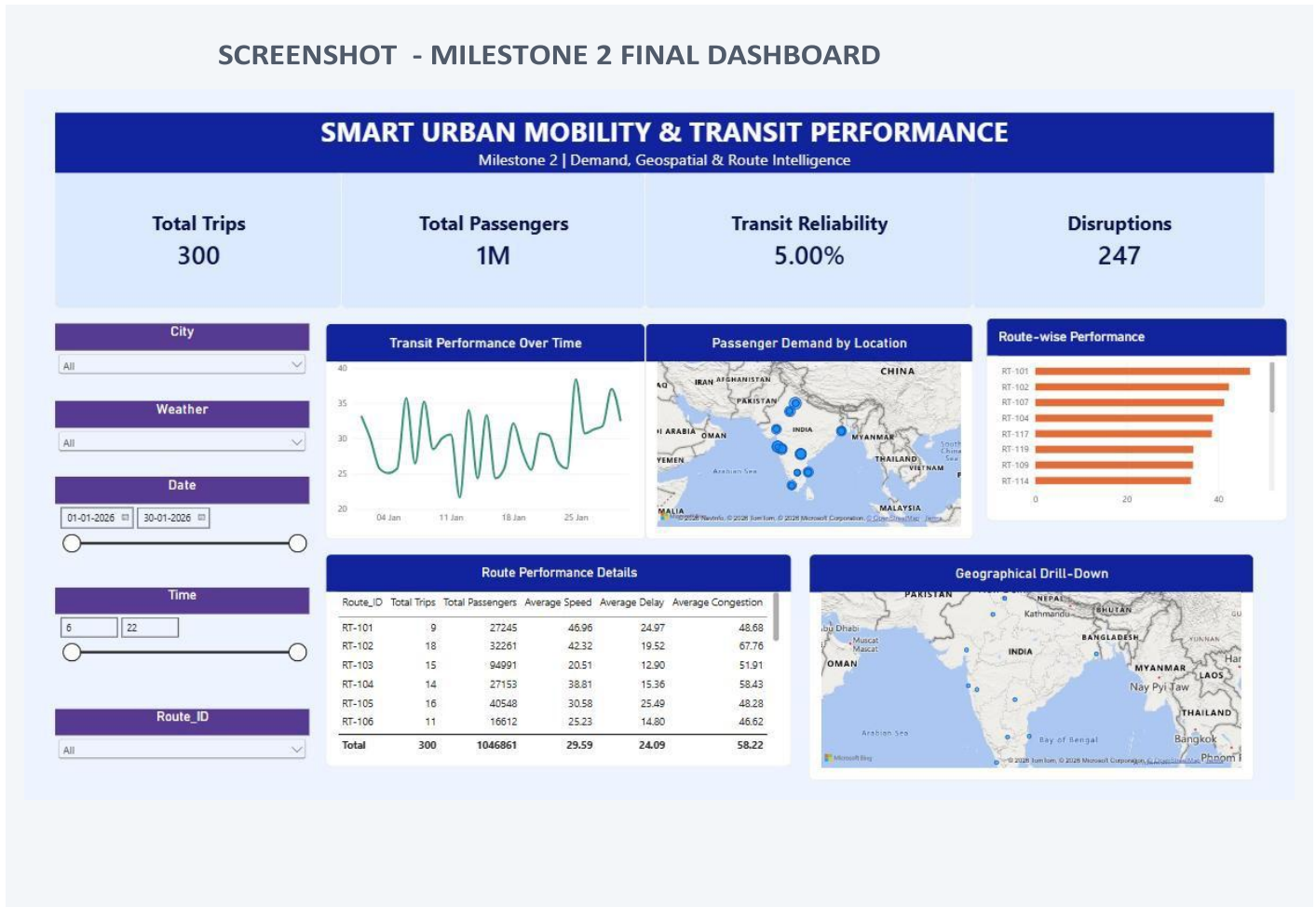


Figure 3. Milestone 2 demand, geospatial and route-intelligence dashboard.

4.5 Origin-Destination Flow Preparation

A MobilityFlow support table was prepared because an origin-destination flow visualization requires ordered geographic points. For each Mobility_ID, the support table stores an origin point and a destination point using Sequence 1 and Sequence 2. Latitude and Longitude are looked up from Dim_Location using LocationKey.

The final interactive Azure Maps flow-path layer could not be displayed in the development environment because the visual requested Microsoft sign-in access. The flow-table preparation and ordered coordinates were retained for later activation.

4.6 Validation and Limitation Notes

- Milestone 1 relationships remained intact before extension.
- Total Trips and Total Passengers responded to route and other slicer selections.
- Transit Reliability was stored as a proportion and displayed as a percentage.
- The route chart, line chart, geographic drill-down and demand map were checked against their intended fields.
- Passenger Satisfaction % is a derived service-quality index and should not be presented as survey data.
- The Azure Maps flow-path visual remained environment-dependent; the support table was prepared instead.

5. Milestone 3 - Demand Intelligence, Mobility Analytics and Forecasting

Milestone 3 moved the project from operational mobility monitoring toward deeper demand intelligence. The same Power BI model and star-schema structure were retained, while demand-focused measures, derived analytical fields, mobility-mode analysis, regional comparison, environmental sensitivity and forecasting exploration were added.

5.1 Data Validation and Basis

The prepared source workbook documents a 300-record mobility sample representing five cities with 60 rows per city, with 0 missing values and 0 duplicate complete rows in the validation record. The sampling method is documented as a stratified random sample with seed 42. The underlying mobility records were retained; fields not directly available in the semantic model were introduced as documented analytical proxies or estimates.

5.2 Derived Analytical Fields

Derived Field	Type / Logic	Interpretation
Estimated Fare	Calculated column: 50 + (12 x route distance in km)	Pricing proxy, not a billed fare.
Average Fare	AVERAGE(Estimated Fare)	Average estimated fare in current filter context.
Surge Multiplier	2.0 if congestion >=80; 1.5 if >=60; 1.25 if >=40; else 1.0	Demand-pressure proxy derived from congestion.
Average Surge	AVERAGE(Surge Multiplier)	Average derived surge multiplier.
Service Provider	Mapped from vehicle categories to analytical provider groups	Provider classification used for demand comparison.
Mobility Mode	Mapped from vehicle categories to Public Transit, Ride-Hailing, Electric Mobility, Private Car and Two-Wheeler groups	Mode classification used for demand composition.
Bike Availability %	Motorcycle trip count / all mobility records	Availability proxy based on motorcycle share.
Region	Mapped directional values to East, South, West and North	Regional grouping for comparison.

5.3 Key DAX Measures

Measure	DAX / Logic	Purpose
Total Demand	SUM(FactMobilityTable[Passenger_Count])	Total passenger demand.
Average Fare	AVERAGE(FactMobilityTable[Estimated Fare])	Average estimated fare.
Average Surge	AVERAGE(FactMobilityTable[Surge Multiplier])	Average congestion-based surge proxy.
Bike Availability %	DIVIDE(CALCULATE(COUNTROWS(FactMobilityTable), Dim_Vehicle[Vehicle_Type] = "Motorcycle"), COUNTROWS(FactMobilityTable), 0)	Motorcycle-share availability proxy.

5.4 Dashboard Visual Specification

Visual	Type	Fields / Configuration	Purpose
Total Demand	Card	Total Demand	Executive demand KPI.
Average Fare	Card	Average Fare	Pricing proxy for current selection.
Average Surge	Card	Average Surge	Demand-pressure pricing proxy.

Demand by Service Provider	Clustered column	Service Provider + Total Demand	Compare demand across modeled provider groups.
Demand by Mobility Mode	Donut	Mobility Mode + Total Demand	Show demand composition across modes.
Temperature vs Demand	Scatter	Temperature_C + Total Demand	Explore temperature-demand relationship.
Demand by Weather	Clustered column	Weather + Total Demand	Compare demand under weather conditions.
Bike Availability % by Region	Clustered column	Region + Bike Availability %	Compare the derived availability proxy across regions.
Demand by Date	Line	Date + Total Demand	Show demand pattern across the reporting period.

5.5 Dashboard Layout and Interaction

The final Milestone 3 page used a professional three-row structure. The top row contained the dashboard title, the three KPI cards - Total Demand, Average Fare and Average Surge - and a Date slicer. The middle row focused on provider demand, mobility-mode composition and the temperature-demand relationship. The bottom row compared demand by weather, bike availability by region and the demand trend over time.

Interactive controls included Date, Region and Mobility Mode. Cross-filtering allowed a selection in one visual to update related visuals through the Power BI semantic model.

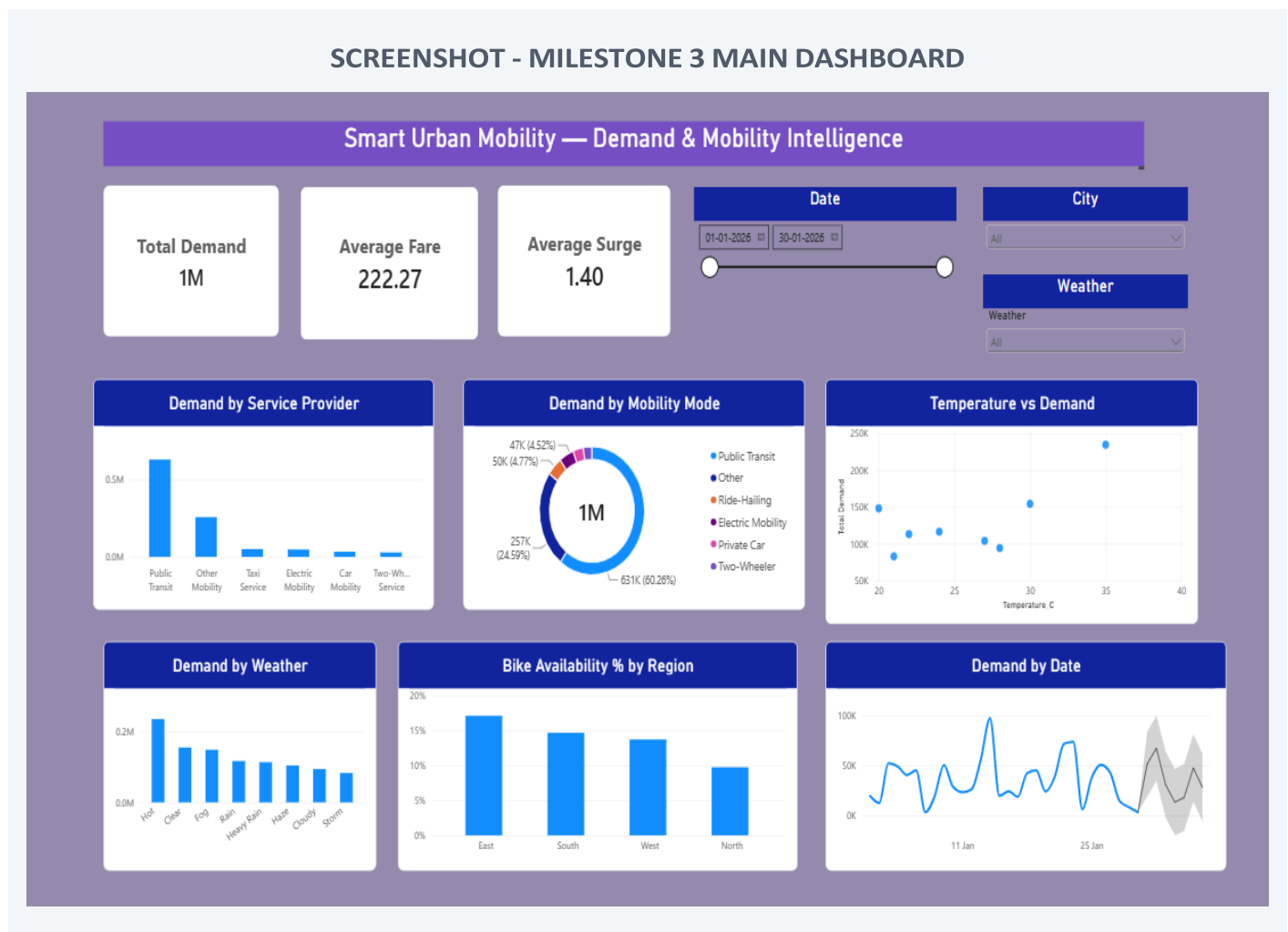


Figure 4. Milestone 3 Smart Urban Mobility - Demand & Mobility Intelligence dashboard.

5.6 Forecasting Exploration

Forecasting was explored on the historical demand trend to extend the dashboard toward forward-looking demand awareness. However, the mentor-specified historical Demand by Date visual remained the primary required visual; forecasting was treated as an enhancement rather than a replacement.

5.7 Transparent Assumptions and Limitations

- Estimated Fare uses a fixed base fare and per-kilometre rate; it is not a billed platform fare.
- Surge Multiplier is derived from Congestion_Index thresholds; it is not an observed platform surge value.
- Service Provider is derived from vehicle categories rather than an original provider/company field.
- Mobility Mode is derived from vehicle categories and is not presented as the source Transport_Mode field.
- Bike Availability is a proxy based on the share of motorcycle records and is not real-time physical inventory.
- Region is derived from Direction categories for the four-region comparison.

6.Milestone 4 Overview and Objectives

Milestone 4 extends the Smart Urban Mobility and Traffic Intelligence Dashboard into an advanced analytical layer while retaining the same Power BI project and underlying mobility model. The milestone focuses on mobility access, mobility equity, executive-level intelligence, demand forecasting and demand anomaly detection.

The work was developed using the existing mobility dataset rather than loading a new dataset. New Milestone 4 indicators are created from available fields using DAX calculated columns and measures.

Objectives

- Create a composite Mobility Access Index using accessibility, reliability and bike-availability components.
- Analyze mobility access across areas and demographic groups through comparative visuals and an equity matrix.
- Create an Executive Intelligence Dashboard with high-level mobility KPIs and supporting analytical visuals.
- Extend historical demand analysis into a Power BI forecasting view.
- Identify unusual demand observations using anomaly detection.
- Present the analysis through an interactive three-page Milestone 4 dashboard.

Milestone 4 Page Structure

Page	Title	Main Content
1	Executive Intelligence	Total Demand, Total Trips, Mobility Access Index, Average Reliability, Average Fare, demand trend, trips by transport mode, travel time by area, demand by location and reliability by mode.
2	Mobility Equity	Mobility Access Index, Average Reliability, Average Travel Time, Total Demand, Average Fare, access by area, demographic comparison, accessibility status and equity matrix.
3	Demand Forecast	Total Demand, historical demand forecast, demand category and demand anomaly detection.

Existing Data and Model Basis

Milestone 4 is built on the established Power BI semantic model from the earlier milestones. The central fact table is FactMobilityTable and it remains connected to Dim_Date, Dim_Location, Dim_Route, Dim_Vehicle and Dim_Weather using the existing one-to-many, single-direction relationships.

The documented project dataset contains 300 mobility records. The location dimension provides geographic attributes such as City, Zone, Location, Latitude, Longitude, Population, Population Density, Road Class and nearby-facility indicators. These existing fields provide the basis for the Milestone 4 access and equity analysis.

Existing Fields Used in Milestone 4

Source Area	Fields Used	Milestone 4 Role
Dim_Date: Date — Time-series and forecasting axis		
Dim_Location: Area, City, Zone, Location, Latitude, Longitude, Near_Hospital, Near_Market, Near_School — Access, equity and geographic analysis		
Dim_Vehicle / FactMobilityTable: Vehicle_Type, Transport_Mode — Mode comparison		
FactMobilityTable: Passenger_Count, Delay_Minutes, Travel_Time_Minutes, Average Fare, Bike Availability — Demand, reliability, travel-time and access indicators		
FactMobilityTable: Demographic_Group, Accessibility_Status, Demand_Category — Milestone 4 segmentation and classification.		

Milestone 4 Analytical Measures and Derived Fields

The Milestone 4 calculations use the existing model and introduce advanced indicators needed by the mentor-specified dashboard. The important principle is that the Mobility Access Index combines three components on a common 0–100 scale: accessibility, reliability and bike availability.

Accessibility Score

Accessibility_Score is derived from the presence of nearby facilities already available in Dim_Location: a hospital, market and school. Each available facility contributes to the score, producing a 0–100 analytical accessibility indicator.

```
Accessibility_Score = ROUND(DIVIDE(Hospital + Market + School, 3) * 100, 2)
```

Accessibility Status

Accessibility_Status classifies the calculated accessibility score into an accessible or underserved category for the accessibility-status distribution visual.

```
Accessibility_Status = IF(Dim_Location[Accessibility_Score] >= 70, "Accessible", "Underserved")
```

Transport Mode

Transport_Mode is a derived analytical grouping created from the available vehicle categories so that the Executive Intelligence Dashboard can compare trips and reliability by transport mode.

```
Transport_Mode = SWITCH(TRUE(), VehicleType="Bus", "Bus", VehicleType="Motorcycle", "Cycling",  
VehicleType="EV", "Metro", VehicleType="Taxi", "Ride-Hailing", VehicleType="Car", "Ride-Hailing", "Other")
```

Demographic Group

Demographic_Group is an analytical grouping created for the mentor-specified equity comparison. It provides the categories used in the dashboard: Student, Senior, Family, Working Age and Low Income.

```
Demographic_Group = SWITCH(Transport_Mode, "Cycling", "Student", "Metro", "Working Age", "Ride-Hailing",  
"Family", "Bus", "Low Income", "Senior")
```

Equity Status

Equity_Status is derived from the accessibility score to support area and demographic equity analysis.

```
Equity_Status = IF(RELATED(Dim_Location[Accessibility_Score]) >= 70, "Equitable", "Needs Attention")
```

Demand Category

Demand_Category groups mobility observations into High, Medium and Low demand using the distribution of Passenger_Count in the current project dataset.

```
Demand_Category = SWITCH(TRUE(), Passenger_Count >= P75, "High", Passenger_Count <= P25, "Low", "Medium")
```

Average Reliability

For Milestone 4 executive analysis, Average Reliability is represented as a 0–100 score derived from the average-delay relationship. This keeps reliability comparable with the other normalized components of the access index.

```
Average Reliability = VAR AvgDelay = AVERAGE(FactMobilityTable[Delay_Minutes])
VAR MaxDelay = MAXX(ALL(FactMobilityTable), FactMobilityTable[Delay_Minutes])
RETURN IF(MaxDelay = 0, 100, (1 - DIVIDE(AvgDelay, MaxDelay, 0)) * 100)
```

Mobility Access Index

The Mobility Access Index is the central Milestone 4 composite measure. It combines the average accessibility score, reliability and bike availability after normalizing reliability and availability to a 0–100 scale.

```
Mobility Access Index = VAR Accessibility = AVERAGE(Dim_Location[Accessibility_Score])
VAR Reliability = [Average Reliability]

VAR BikeAvailability = [Bike Availability % Measure] * 100
RETURN DIVIDE(Accessibility + Reliability + BikeAvailability, 3)
```

A higher index represents a higher combined value across the three analytical components; a lower index indicates that one or more components contribute a lower score. The measure is intended for comparison under the active Power BI filter context.

Page 1 – Executive Intelligence Dashboard

The first Milestone 4 page is the Executive Intelligence Dashboard. It provides a high-level summary of mobility demand, trip activity, access, reliability and fare, followed by supporting visuals for trends, transport modes, area travel time, geographic demand and mode reliability.

KPI Cards

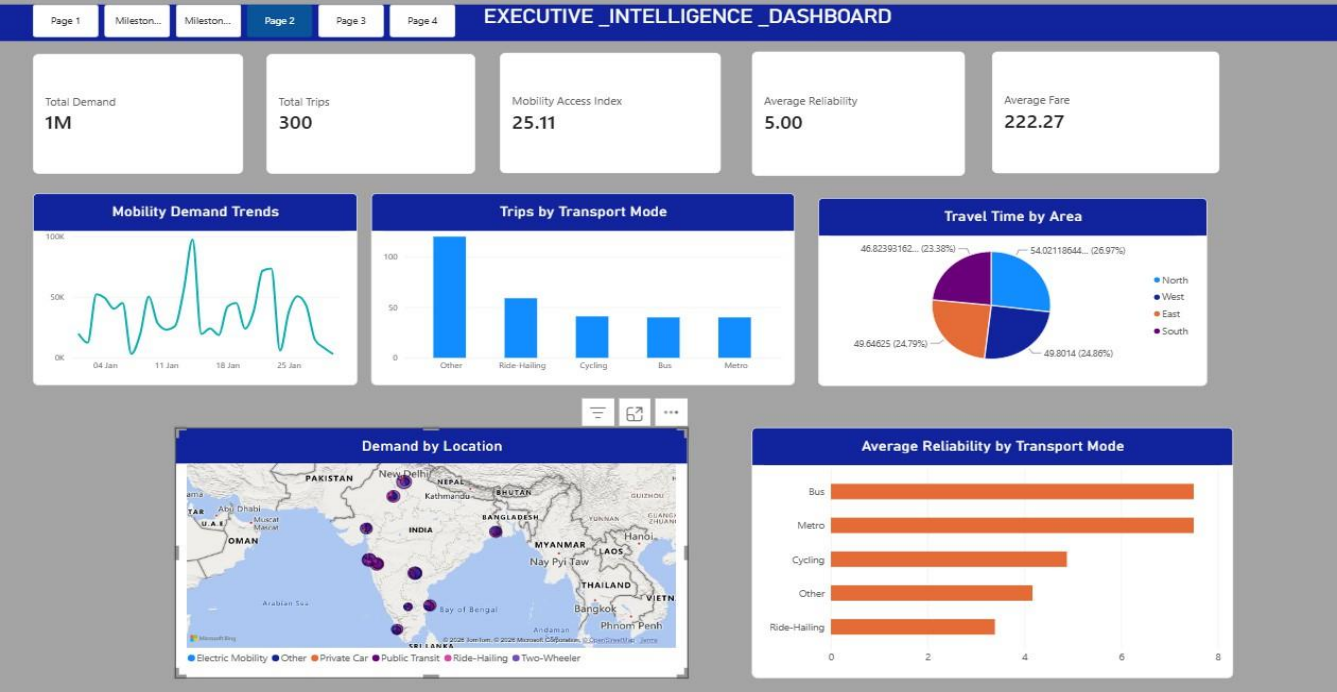
- Total Demand
- Total Trips
- Mobility Access Index
- Average Reliability
- Average Fare

Supporting Visuals

Visual	Type	Purpose
Mobility Demand Trends	Line chart	Show demand variation over date.
Trips by Transport Mode	Clustered column chart	Compare trip volume across transport modes.
Travel Time by Area	Donut/Pie chart	Compare travel-time distribution across areas.
Demand by Location	Map	Show geographic concentration of demand.
Average Reliability by Transport Mode	Bar chart	Compare reliability across transport modes.

Executive Interpretation

The page is designed as a top-to-bottom executive view. The KPI cards establish the current mobility state, the trend visual shows how demand changes over time, the mode chart shows trip composition, the area chart compares travel-time distribution, the map shows geographic demand, and the reliability chart compares service performance across transport modes.



Page 2 – Mobility Equity and Access Index

The second page focuses on mobility equity. It uses the Mobility Access Index as a common metric and compares access across areas and demographic groups while retaining the same project data and filter context.

KPI Cards

- Mobility Access Index
- Average Reliability
- Average Travel Time
- Total Demand
- Average Fare

Equity Visuals

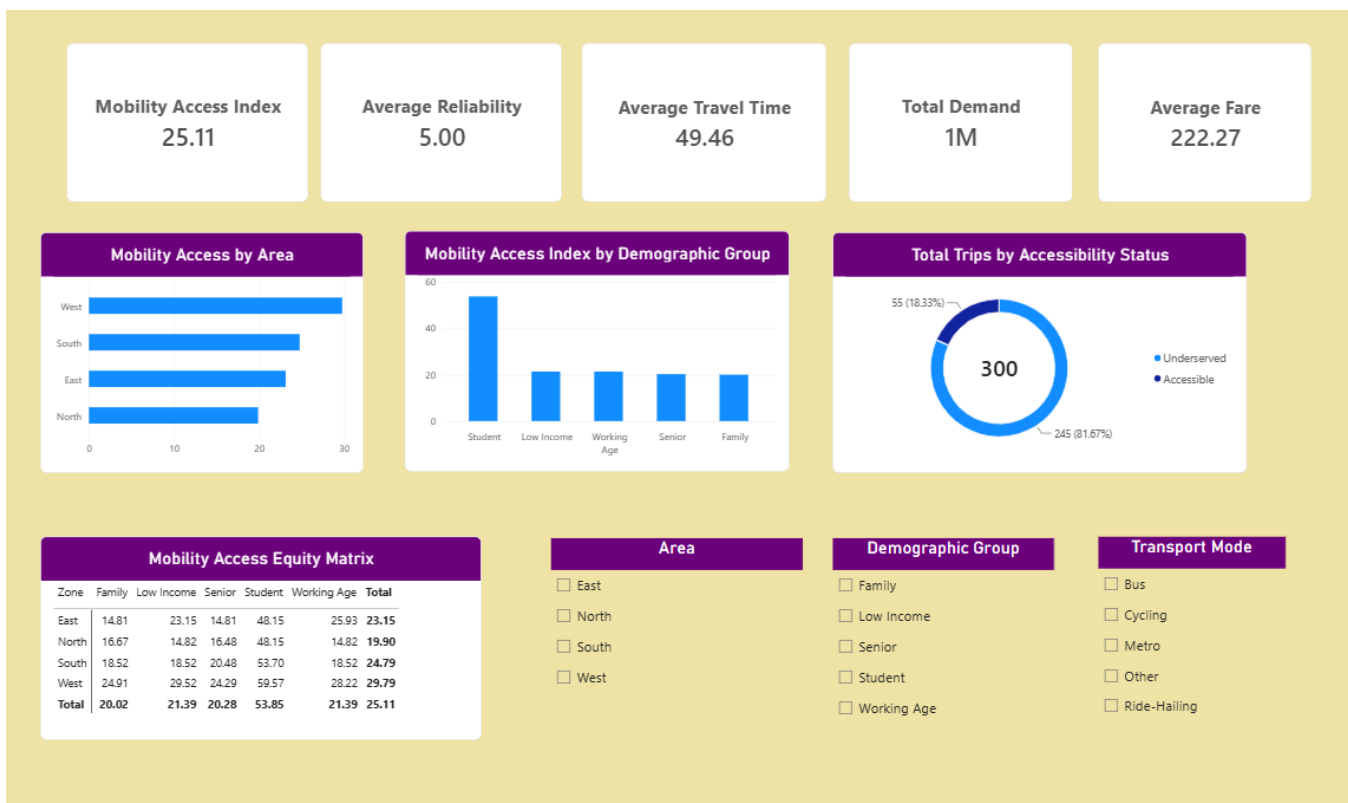
Visual	Type	Purpose
Mobility Access by Area	Bar chart	Compare Mobility Access Index across areas.
Mobility Access Index by Demographic Group	Column chart	Compare access across the mentor-specified demographic groups.
Total Trips by Accessibility Status	Donut chart	Show the distribution of trips by accessibility status.

Filters

- Area
- Demographic Group
- Transport Mode

Equity Interpretation

The equity page supports comparison rather than a single aggregate view. Area-level access can be examined alongside demographic-group values, while the matrix provides a compact area-by-group view. The filters allow the analysis to be narrowed to a selected area, demographic group or transport mode.



Page 3 – Demand Forecasting and Anomaly Detection

The third page extends the dashboard from descriptive demand reporting into forward-looking analysis. The page keeps Total Demand as the headline KPI, uses historical demand as the forecasting basis, classifies observations by demand category and highlights unusual demand behavior through anomaly detection.

Headline KPI

- Total Demand

Historical Demand Forecast

A line chart uses Dim_Date[Date] on the horizontal axis and Total Demand on the vertical axis. Power BI forecasting is applied to the historical series to produce a forward-looking forecast with a confidence interval. Forecast setting used during the implementation: forecast length of 30 periods and a 95% confidence interval.

Demand Category

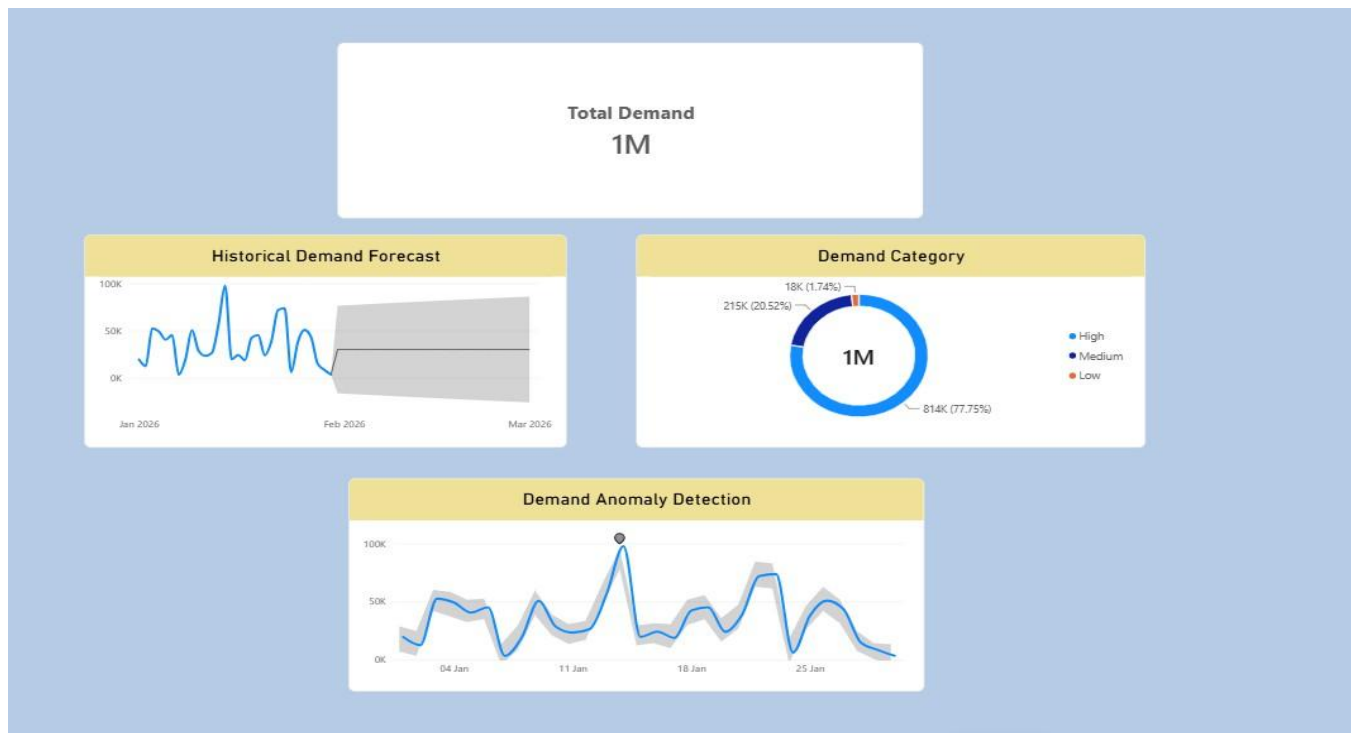
The Demand Category visual summarizes High, Medium and Low demand observations. This gives the executive page a simple classification layer alongside the continuous time-series view.

Demand Anomaly Detection

Anomaly detection is applied to the historical Total Demand line chart. Power BI identifies observations that depart unusually from the historical pattern, allowing spikes or drops to be highlighted without replacing the underlying demand trend.

Page 3 Analytical Workflow

- Read the Total Demand KPI.
- Inspect the historical demand trend.
- Use the forecast band to examine forward-looking demand behavior.
- Use Demand Category to understand the distribution of demand levels.
- Use anomaly markers to identify unusual demand observations.



Interaction and Validation

Milestone 4 continues the interactive behavior established in the earlier milestones. Measures recalculate in the current filter context, allowing the same dashboard to be examined by different areas, demographic groups, transport modes and dates.

Validation Checks

- Confirmed that the existing fact/dimension model remains the analytical foundation.
- Checked that KPI measures and derived indicators populate under the current filter context.
- Validated area and demographic-group comparisons in the equity page.
- Checked that the forecasting visual uses the date-based historical demand series.
- Applied anomaly detection to the historical demand series.
- Checked that the three Milestone 4 pages retain a consistent visual structure.

User Workflow

The intended workflow is: begin with executive KPIs, compare overall access and equity, then move to historical demand, forecast behavior and anomaly detection. This provides a progression from summary indicators to more detailed analytical evidence.

Assumptions and Analytical Limitations

Milestone 4 uses derived analytical fields where equivalent source fields are not directly available in the original mobility dataset. These fields are useful for dashboard analysis but should be interpreted as analytical constructs rather than direct survey or real-time operational measurements.

- Accessibility_Score is derived from nearby-facility indicators available in Dim_Location.
- Demographic_Group is an analytical grouping created for the mentor-specified equity analysis; it is not a direct demographic survey field in the source dataset.
- Accessibility_Status and Equity_Status are classification fields derived from the calculated access score.
- Mobility Access Index is a composite analytical index combining accessibility, reliability and bike availability.
- Bike availability remains the previously defined proxy based on the motorcycle-share measure; it is not real-time physical inventory.
- Demand forecasting is based on the historical demand series available in the project dataset.
- Demand anomaly detection identifies unusual observations relative to the historical series and should be interpreted within the available sample and time range.
- The Demand by Location map may remain environment-dependent in Power BI Desktop if the map visual/service is unavailable; the analytical intent and coordinate fields remain part of the design.

7. Conclusion

Milestone 4 extends the Smart Urban Mobility and Traffic Intelligence Dashboard into advanced analytics using the same Power BI model established in the earlier milestones. The milestone combines mobility access, equity analysis, executive intelligence, forecasting and anomaly detection within a structured three-page dashboard.

The Mobility Access Index provides a single composite indicator from accessibility, reliability and bike availability. The equity page enables area and demographic comparison, the executive page summarizes core mobility performance, and the forecasting page adds a forward-looking demand view together with anomaly detection.

Overall, Milestone 4 demonstrates how the existing mobility data model can be extended with DAX-based analytical indicators and Power BI advanced visuals to support a richer decision-support workflow.

8.References

- Microsoft Power BI Desktop documentation and visualization guidance.
- Microsoft DAX reference and calculation concepts.
- Microsoft Power Query documentation.
- Project Smart Urban Mobility dataset and validation workbook.
- Infosys Springboard Smart Urban Mobility project milestone documentation.
- Mentor-provided Milestone 4 dashboard structure and meeting walkthrough.

